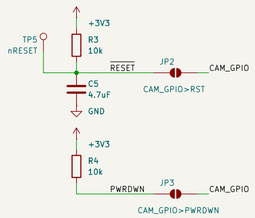
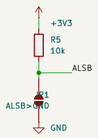
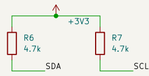
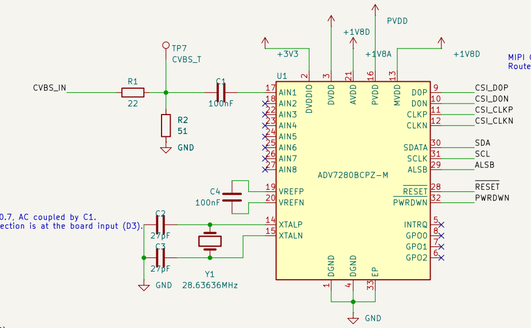
Reset / power-down / address
Overlay adv728x-m sets no reset/powerdown GPIO and does not enable cam1_reg,
so RESET and PWRDWN are pulled up on board; JP2/JP3 optionally route CAM_GPIO (Pi pin 11).
ALSB=1 -> 7-bit addr 0x21 as in adv7282m-overlay.dts (reg = <0x21>). Close JP1 for 0x20.





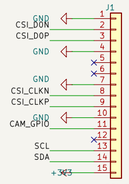


MIPI CSI-2, 1 data lane + clock (216 Mbps interlaced / 432 Mbps I2P).
Route as 100 R differential (2 x 50 R loosely coupled per UG-637), GND plane under; P/N crossover: both lines 2 vias + GND return vias.



Analog input (UG-637 Fig. 9): 75 R termination (R1+R2), gain 0.7, AC coupled by C1.
CVBS_IN is driven by THS7314 CH1 through R9 (75 R); ESD protection is at the board input (D3).

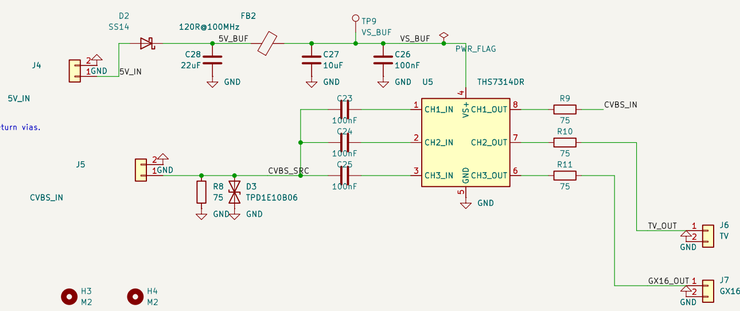
Crystal 28.63636 MHz, fundamental, CL 20 pF:
C = 2(CL-Cs)-Cpg = 2(20-3)-4 = 30 pF -> 27 pF (AN-1260)



Pi 4 J3 (CAMERA) pinout, 1:1 with Camera Module v2.1:
1 GND 2 D0N 3 D0P 4 GND 5 D1N(nc) 6 D1P(nc) 7 GND
8 CLKN 9 CLKP 10 GND 11 CAM_GPIO 12 nc 13 SCL0 14 SDA0 15 3V3
I2C on Pi 4 = GPIO44/45 (I2c_csi_dsi); pull-ups to 3V3 exist on the Pi.

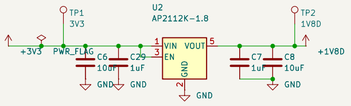
THS7314 BUFFER STAGE (rev C: one board for the SPRUT-2 case)

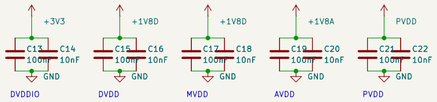
5 V from the case DC-DC 5 V module via J4 (JST-XH): the Pi 4 is powered from the same module. The decoder itself runs from the Pi 3.3 V (J1 pin 15).
Panel BNC -> coax -> J5 -> buffer: CH1 -> decoder, CH2 -> J6 TV (AV in), CH3 -> J7 GX16 monitor.



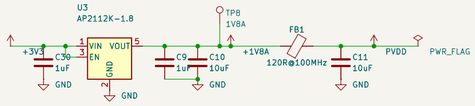
Power: J4 -> D2 SS14 (reverse polarity, ~0.4 V) -> C28 22 uF -> FB2 -> C27 10 uF + C26 100 nF at VS+ (SLOS513A: 0.1 uF at the pin, 22-100 uF on the line).
THS7314 runs from 3 to 5 V; at ~4.6 V the output swing (4.5 V typ into 75 R) leaves >2 V margin for 2 Vpp video + 0.29 V clamp offset. No LDO needed. Iq 17 mA.

Input (SLOS513A Fig. 33/34): R8 75 R termination, D3 ESD, C23-C25 100 nF series -> sync-tip clamp (DC restore) mode, one cap per input (each input has its own clamp).
Outputs: gain 6 dB (x2), series 75 R into 75 R load -> unity overall. CH1 -> decoder (R1+R2 = 73 R load), CH2/CH3 -> J6 TV / J7 GX16. Bandwidth 8.5 MHz (5th-order LPF).





Decoupling per UG-637: 100 nF + 10 nF at each supply pin, same side as U1, via under the 100 nF pad.



Power: 3.3 V from Pi camera connector (pin 15). Two AP2112K-1.8 LDOs:
U2 -> DVDD+MVDD (~84 mA typ). U3 -> AVDD (~47 mA) and PVDD via FB1 (~12 mA).
UG-637 recommends separate regulated supplies for the analog/PLL group.



Sources: ADV7280 DS Rev.A, UG-637 Rev.A, AN-1260; see docs/requirements.md
Target: Raspberry Pi 4 CSI camera port, dtoverlay=adv728x-m,adv7280m=1

Sheet: /
File: adv7280m-csi2.kicad_sch

Title: CVBS to MIPI CSI-2 decoder ADV7280-M, Pi 4, SPRUT-2

Size: A2	Date: 2026-10-15	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/1